Software Architecture Styles & Design Patterns

RCS 305 ? Software Engineering | Riara University Revision Resource

1. Architectural Styles

? Layered (N-Tier): Segregates Presentation, Business Logic, and Data Persistence layers. Changes in one layer do not propagate if interfaces remain constant.
? Microservices: Decomposes monolithic applications into independently deployable, loosely coupled services communicating via HTTP/REST or message queues.
? Event-Driven: Decouples event producers from consumers via pub/sub brokers (e.g., Kafka, RabbitMQ) for massive horizontal scalability.

2. Gang of Four (GoF) Design Patterns

? Creational ? Singleton: Guarantees a class has only one instance with a global access point.
? Creational ? Factory Method: Encapsulates object instantiation behind an interface, letting subclasses decide which class to instantiate.
? Structural ? Adapter: Converts the interface of a class into another interface clients expect.
? Behavioral ? Observer: Defines a one-to-many dependency so that when one object changes state, all dependents are notified automatically.

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